

SemEval 2026 Task 4: Narrative Similarity

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Abstract

This report details our methodological advancements and experimental findings for Milestone 2 of the SemEval Narrative Similarity task. To address the scarcity of human-annotated data and mitigate vocabulary matching biases in dense encoders, we developed a large-scale synthetic data augmentation pipeline using GPT-5.4-mini to generate procedurally diverse narrative triplets. This was paired with a Theme-Action-Outcome (TAO) decomposition strategy designed to strip away surface-level tropes and extract pure narrative skeletons. However, fine-tuning dense retrieval models on this skeletonized data resulted in unexpected performance degradation, as the models struggled with severe information loss and out-of-distribution syntactic formatting. Consequently, we pivoted to an LLM-as-a-Judge paradigm and leveraged a few-shot structural reasoning. Extensive evaluations across local open-source architectures and frontier APIs revealed a stark capability divide, as frontier-class generative models (GPT-4o) successfully decoupled structural alignment from semantic noise, establishing a new baseline accuracy of 0.755.

1 Dataset Augmentation

To address the challenge of limited training data for the SemEval Narrative Similarity task, we implemented a synthetic data augmentation pipeline. By leveraging a large language model, we procedurally generated a diverse, large-scale dataset of narrative triplets designed to mirror the structural nuances of the target task. This approach allowed us to introduce controlled variability in genre, pacing, and narrative structure, ultimately forcing downstream models to learn robust representations of abstract plot mechanics rather than relying on superficial textual overlaps or length-based heuristics.

The augmented dataset was generated using the following methodology:

- **Model and Architecture:** We utilized OpenAI’s gpt-5.4-mini model, leveraging its Structured Outputs feature (enforced via Pydantic schemas) to guarantee perfectly formatted JSON batches containing Story Triplets (Anchor, Similar Story, and Dissimilar Story).
- **Dynamic Constraint Injection:** To prevent the model from collapsing into repetitive narrative patterns, each generated triplet was heavily parameterized. We randomly sampled combinations of 8 media types (e.g., feature films, stage plays), 17 specific genres (e.g., Action Movies, Gothic Horror), and 11 distinct conflict arcs to serve as the foundational prompt for each text.
- **Stylistic and Structural Diversity:** The prompt enforced distinct structural variations for the summaries. Styles ranged from chronological scene-by-scene breakdowns to non-linear academic retrospectives and actor-heavy synopses. We also injected randomly selected mandatory opening hooks to vary the starting syntax.
- **Length Jittering:** To prevent downstream models from using text length as a lazy classification heuristic, we enforced strict target lengths for the generated summaries. These targets varied across five tiers, ranging from brief 30-word synopses to dense 180-word descriptions.
- **Multi-Track Formatting:** The generated triplets were automatically parsed into two distinct dataset formats to support different evaluation tracks. One format retained the raw triplet structure for contrastive learning, while the other randomly shuffled the similar and dissimilar stories into Candidate A and Candidate B to provide balanced, binary classification targets for our LLM-as-a-Judge pipeline.

2 Theme-Action-Outcome (TAO) Decomposition

To mitigate the risk of models relying on superficial semantic overlaps such as shared genres, character names, or exotic settings, we introduced a Theme-Action-Outcome (TAO) decomposition step. This preprocessing pipeline acts as a “narrative skeletonization” process, systematically stripping away distracting surface-level tropes to isolate the pure structural mechanics of each story. By reducing complex narrative summaries into strict, abstract symbolic triples, we compel downstream embedding models and evaluators to focus exclusively on true structural alignment rather than vocabulary-level vibe-matching.

The TAO extraction dataset was generated using the following methodology:

- **Extraction Engine and Schema:** We utilized the gpt-4o-mini model operating at a low temperature (0.1) to ensure highly deterministic and objective deconstruction. Using OpenAI’s Structured Outputs, we enforced a strict Pydantic schema to parse each story into three distinct variables: Theme, Action, and Outcome.
- **Entity Sanitization:** To neutralize semantic biases, the prompt enforced a “No Proper Nouns” rule. The model was instructed to strip out all specific character names, temporal settings, locations, and unique items, replacing them with generic, archetypal identifiers (e.g., “the protagonist”, “an artifact”, “an isolated location”).
- **Relational Integrity:** The extraction engine processed the stories in their original triplet format (Anchor, Similar, Dissimilar) to maintain contextual boundaries. The prompt explicitly mandated that the similar story’s TAO must closely mirror the anchor, while the dissimilar story must reflect a fundamentally different action or outcome.
- **Semantic Flattening for Embeddings:** Rather than storing the output as nested JSON, the parsed TAO components were concatenated back into a dense, flat string (e.g., Theme: [...] Action: [...] Outcome: [...]). This specific formatting was designed to provide highly semantic, information-dense text sequences optimized for downstream contrastive learning with dual-encoder embedding models.

The figures below show how a single triplet from the original dataset is modified to comply to the Theme-Action-Outcome schema:

Raw dataset sample (Extracted from our synthetic Dataset)

```
{
  "anchor_story": "The story revolves around Mateo Rivas (Clint Eastwood),
    the owner of a dusty border town that depends on a shallow river and a
    failing well. When surveyors announce that a seasonal sinkhole has
    opened beneath the aquifer, the townspeople begin rationing water and
    moving livestock to higher ground. [...]",
  "similar_story": "The story revolves around El\u00edas Navarro (Javier
    Bardem), the proprietor of a plateau settlement threatened by a slow
    volcanic ashfall that poisons wells and blurs the horizon. [...]",
  "dissimilar_story": "The story revolves around Mateo Rivas (Clint Eastwood
    ), who uses the sinkhole scare to stage property seizures, close the
    saloon, and force the widow and teacher to accept his private control
    of the remaining water. The river never fully fails, and the freight
    train delivers enough supplies for the town to stay in place. [...]"
}
```

After we run the TAO processing pipeline, we get the following result:

TAO Result for this sample

```
{
  "anchor_story": "Theme: Environmental degradation leads to community collapse Action: Residents exploit natural resources until disaster strikes Outcome: Community abandons land, allowing nature to reclaim it.",
  "similar_story": "Theme: Environmental degradation leads to community collapse Action: Residents exploit natural resources until disaster strikes Outcome: Community abandons land, allowing nature to reclaim it.",
  "dissimilar_story": "Theme: Environmental disruption leads to community upheaval Action: Residents seek new resources amidst strange phenomena Outcome: Community evacuates, leading to a transformed environment."
}
```

3 Experiments

To address the limitations of standard embedding models—specifically their vulnerability to superficial semantic overlaps and the scarcity of human-annotated training data—we evaluated the impact of our Theme-Action-Outcome (TAO) decomposition and our synthetic augmented dataset.

We tested these interventions using the bge-large-en-v1.5 architecture, which had previously established a strong zero-shot baseline of 0.680 on Track A and 0.652 on Track B. The results of our experiments are presented in Table 1:

Model	Param	Track A (Acc)	Track B (Acc)
bge-large-en-v1.5	330M	0.680	0.652
all-MiniLM-L6-v2 (Tuned)	22M	0.662	0.592
Qwen-2.5-7B-instruct	7B	0.560	-
nli-deberta-v3-large	304M	0.520	-
uhhlt/story-emb	7B	0.520	-
all-MiniLM-L6-v2 (Zero-shot)	22M	0.500	-
deberta-v3-large	304M	0.490	-
Jaccard (Baseline)	-	0.560	-
bge-large-en-v1.5 + Augmented Dataset	330M	-	0.60
TAO + bge-large-en-v1.5	330M	0.61	0.59
TAO + Augmented Dataset + bge-large-en-v1.5	330M	0.63	0.625

Table 1: Performance comparison of baseline, cross-encoder, contrastively fine-tuned models from Milestone 1, as well as the newly trained models. Track B requires standalone vector representations, prohibiting cross-encoder and direct LLM evaluation.

Surprisingly, the experimental results indicated a consistent degradation in performance. Fine-tuning the model on the raw augmented dataset dropped accuracy to 0.620 (Track A) and 0.600 (Track B). Applying the zero-shot TAO decomposition without fine-tuning yielded the lowest scores at 0.610 (Track A) and 0.590 (Track B). Finally, combining both approaches: fine-tuning the model explicitly on TAO-formatted augmented data, along with the existing data resulted in a slight recovery to 0.630 (Track A) and 0.625 (Track B), though it still failed to surpass the raw, zero-shot baseline.

An analysis of these regressions reveals several critical insights into how dense retrieval models interact with heavily pre-processed and synthetically generated text:

- **Catastrophic Information Loss:** The TAO pipeline acts as a highly lossy compression algorithm. While stripping away proper nouns, genres, and settings successfully removes superficial vibe biases, it also destroys the rich, latent contextual fabric that contrastive models like BGE rely on to map semantic space. By reducing a complex narrative into a rigid, three-variable skeleton, we inadvertently starved the embedding model of the linguistic nuances necessary to distinguish hard negatives.
- **Out-of-Distribution Formatting:** Dense embedding models are pre-trained on fluid, natural language prose (e.g., Wikipedia, Reddit, web crawl data). The strict, concatenated string format generated by the TAO pipeline (Theme: [...] Action: [...] Outcome: [...]) is highly artificial. Consequently, the zero-shot TAO + BGE pipeline performed the worst, as the model struggled to accurately embed a syntactic structure it had rarely encountered during pre-training.
- **The Synthetic Distribution Trap:** The degradation observed when fine-tuning on the augmented dataset (without TAO) suggests that the model overfit to the synthetic patterns generated by the LLM. Rather than learning a deeper understanding of narratology, the BGE model likely learned to identify the superficial heuristics and pacing regularities inherent to the GPT-generated text, failing to generalize to the human-authored complexities of the SemEval test set.

4 LLM-as-a-Judge

Traditional dense embedding models and cross-encoders frequently struggle with the deep, abstract reasoning required for structural narratology. By collapsing complex event sequences into dense mathematical vectors or relying on token-level attention, these models often lose the mechanical nuance of a story’s course of action and abstract theme. Conversely, LLMs possess massive context windows and advanced zero-shot reasoning capabilities.

We initially hypothesized that framing the SemEval task as an LLM-as-a-Judge problem would allow a model to simultaneously hold the Anchor, Candidate A, and Candidate B in its memory, explicitly reasoning through the grading rubric to separate underlying plot mechanics from superficial semantic overlaps. By prompting the LLM to map the narrative skeleton before making a decision, we expected to bypass the vocabulary-matching pitfalls that hindered traditional encoders.

For our experiments, we prompted multiple LLMs, both open and closed-source, and of varying sizes. The experiments were mostly done with few-shot prompting using the human-annotated samples provided by the competition. We also tried a zero-shot approach, as well as a majority voting (self-sufficiency) approach. The master prompt used for all models is shown below and was designed to create a Chain-of-Thought reasoning within the LLMs:

LLM Master Prompt

```
You are an expert narratologist grading the SemEval Narrative Similarity
task.
Your job is to determine whether TEXT A or TEXT B is structurally closer to
the ANCHOR.

SEMEVAL ANNOTATION GUIDELINES:
1. Ignore superficial genres, character names, and settings.
2. 'Course of Action' (the mechanical steps the characters take) is heavily
   weighted.
3. 'Abstract Theme' (the philosophical core) is heavily weighted.
4. 'Outcome' differences do NOT automatically disqualify a text if the
   Course of Action matches.

Think step-by-step. Map the narrative mechanics of Anchor vs A, then Anchor
vs B. Then declare the winner.

CRITICAL: You MUST output ONLY a valid JSON object. Do not include any
markdown formatting, backticks, greetings, or preamble.
```

The results from the LLM-as-a-Judge experiments, along with the best previous result, are shown in Table 2 below, indicating a clear winner. Since we are using LLMs, we can only evaluate the outputs on Track A. The official winner of the competition scored a 0.78 score using a supermajority vote with undisclosed LLMs and symbolic methods for tie breaking.

Model	Param (Acc)	Track A
OpenAI GPT-4o (Few-Shot)	1.8T	0.7550
Deepseek-Reasoner (Few-Shot)	671B	0.7225
OpenAI GPT-5.2 (Few-Shot)	<i>n/a</i>	0.7220
Meta Llama 3.1 (Few-Shot + Majority Vote)	70B	0.6820
OpenAI GPT-4o-mini (Few-Shot)	8B	0.6800
bge-large-en-v1.5	330M	0.6800
Meta Llama 3.1 (Few-Shot)	8B	0.6600
OpenAI GPT-4o-mini (Zero-Shot)	8B	0.6550

Table 2: Performance comparison of the LLM models, as well as the previous best model.

Evaluating the models across a spectrum of parameter scales revealed a stark divide between frontier-class models and mid-tier generative architectures. The results demonstrate that while LLMs can perform abstract structural mapping, success is heavily gated by model scale.

- **The Frontier Model Advantage:** OpenAI’s GPT-4o emerged as the definitive top performer, achieving an accuracy of 0.755. This significant leap over the dense retrieval baseline (BGE-large at 0.680) suggests that at the 1.8T parameter scale, the model possesses the requisite abstraction capabilities to successfully decouple a story’s underlying mechanics from its superficial semantic “vibe”.
- **The Overthinking Trap in Reasoning Models:** Surprisingly, the dedicated reasoning models—Deepseek-Reasoner (0.7225) and the newly tested GPT-5.2 (0.722) failed to surpass GPT-4o, despite their advanced chain-of-thought architectures. This indicates that forcing an extensive, multi-step internal monologue may occasionally cause the model to over-analyze minor narrative discrepancies, losing sight of the broader structural alignment that standard frontier models map intuitively.
- **The Mid-Tier Plateau:** The results explicitly confirm the local open-source ceiling hypothesized earlier. Meta Llama 3.1 70B, even when augmented with a rigorous Few-Shot and Majority Vote pipeline, peaked at 0.682. Similarly, OpenAI’s GPT-4o-mini (Few-Shot) scored 0.680. Both models effectively tied with the much smaller bge-large-en-v1.5 dense encoder (0.680), proving that simply deploying a conversational LLM does not automatically grant superior structural understanding unless the model operates at the absolute frontier scale.
- **The Necessity of In-Context Learning:** The ablation between GPT-4o-mini’s Few-Shot (0.680) and Zero-Shot (0.655) performance highlights the critical role of in-context anchoring. Without explicit examples of the grading rubric in the prompt, generative models rapidly regress, likely falling back into standard semantic vocabulary-matching behaviors.
- **The Sub-10B Parameter Collapse:** The smallest models tested, including Meta Llama 3.1 8B (0.660) and the Zero-Shot GPT-4o-mini (0.655), performed worse than the 330M parameter embedding baseline. This definitively shows that smaller generative models lack the cognitive span and attention mechanisms required to simultaneously hold three complex stories in their context window and execute an abstract comparative analysis without crossing narrative wires.

5 Conclusion

In conclusion, our exhaustive exploration of the SemEval Narrative Similarity task highlights the profound difficulty of teaching machines to decouple abstract plot mechanics from superficial semantic details. Traditional dense retrieval models and fine-tuned cross-encoders, even when supported by extensive synthetic data augmentation and strict TAO (Theme-Action-Outcome) decomposition, proved fundamentally vulnerable to information loss and synthetic data overfitting. Similarly, local generative architectures, including highly optimized 70-billion parameter models and advanced chain-of-thought reasoning engines hit a hard performance ceiling.

These mid-tier models consistently struggled with attention dilution and the vibe check trap, failing to maintain rigid structural boundaries when holding multiple complex narratives in context. Ultimately, our empirical results demonstrate that mastering computational narratology requires a level of zero-shot deductive abstraction currently exclusive to frontier-class models. By pivoting from local pipelines to the massive scale and adaptive reasoning capabilities of the Deepseek, GPT-4 and GPT-5 architectures, we established a state-of-the-art baseline of 0.755, proving that the future of structural narrative alignment lies not in heavily pre-processed embeddings, but in raw, frontier-level cognitive scale.